Chapter 7: Proportional Reasoning – 1 (Ratio, Proportion, and Percentages)

Worksheet 7: Ratio, Proportion, and Percentages

Learning Objective: Students will work with ratios and proportions; solve direct and inverse proportion problems; apply percentages to profit, loss, simple interest, and discount; and tackle compound interest problems.

Worked Example

Simple Interest: $SI = \dfrac{P \times R \times T}{100}$

Example: $P = \text{Rs.}\,8000$, $R = 12\%$, $T = 3$ years

$SI = \dfrac{8000 \times 12 \times 3}{100} = \text{Rs.}\,2880$; Amount $= \text{Rs.}\,10880$

$\text{Profit\%} = \dfrac{\text{Profit}}{\text{CP}} \times 100$; if CP $= \text{Rs.}\,500$ and SP $= \text{Rs.}\,575$: Profit% $= 15\%$.

Questions 1–5: Easy / Recall

Q1. Simplify the ratio: $48 : 36$.

Q2. A map has scale $1 : 50000$. A road is 5 cm on the map. Find its actual length.

Q3. A shopkeeper buys at Rs. 400 and sells at Rs. 460. Find profit%.

Q4. Find the SI on Rs. 5000 at 8% per annum for 2 years.

Q5. x and y are directly proportional. If $x = 6$ when $y = 18$, find y when $x = 10$.

Questions 6–10: Basic Application

Q6. A shopkeeper marks a shirt at Rs. 800 and gives 15% discount. Find the selling price.

Q7. If 8 workers can complete a job in 12 days, how many workers are needed to finish it in 6 days? (Inverse proportion.)

Q8. Find CI on Rs. 10000 at 10% per annum for 2 years (compounded annually).

Q9. A train travels 300 km in 4 hours. At the same speed, how far does it travel in 7 hours?

Q10. A loss of 12.5% is made on an item sold for Rs. 875. Find the cost price.

Questions 11–15: Practice and Skill Building

Q11. The ratio of boys to girls in a school is 5 : 4. If there are 540 students, find the number of boys and girls.

Q12. A car depreciates at 10% per year. It costs Rs. 4,00,000 now. Find its value after 2 years.

Q13. Population of a town is 80,000. It grows at 5% per year. Find population after 2 years (using CI formula).

Q14. At what rate of simple interest will Rs. 1200 double itself in 8 years?

Q15. Three quantities A, B, C are in the ratio $2 : 3 : 5$. Their sum is 1200. Find each.

Questions 16–18: Higher Order Thinking

Q16. A sum of money becomes Rs. 6050 after 2 years at 10% CI. Find the principal.

Q17. A shopkeeper gives two successive discounts of 20% and 10%. What single equivalent discount does this represent?

Q18. Ramu sold two articles for Rs. 990 each. On one he gained 10% and on the other he lost 10%. Find his overall profit or loss and the percentage.

Questions 19–20: Real-Life Applications

Q19. Ayesha deposits Rs. 25,000 in a bank at 8% CI per annum. Find the amount after 3 years. How much interest does she earn?

Q20. During a sale, a mobile phone is available at 20% discount and an additional 5% off on the discounted price. If the marked price is Rs. 18,000, find the final price and total savings.

Reflection Question: What is the difference between simple interest and compound interest? Why does compound interest grow faster? Use a table to compare for Rs. 10,000 at 10% for 3 years.